MicroViT: Global Self-Attention, INT8 Edge Quantization, and Online Temporal Filtering for High-Speed Autonomous Scale Racing Under Severe Environmental Drift

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Abstract—Autonomous scale-vehicle racing at speeds exceeding 20 km/h (> 5.5 m/s) imposes strict physical deadlines, requiring deterministic steering decisions within 10 ms (≤ 5 cm of track displacement) under sub-watt edge compute budgets. While classical Convolutional Neural Networks (CNNs) such as NVIDIA DAVE-2 and the ENSICAEN baseline achieve low nominal inference latency, their localized receptive fields suffer catastrophic performance breakdown under environmental domain shifts—such as sharp barrier shadows, track-side spectator occlusions, and venue illumination drops. In this paper, we present **MicroViT**, an ultra-compact Vision Transformer engineered specifically for edge autonomous racing, paired with an INT8 TensorRT/ONNX optimization pipeline and a real-time causal temporal filter. Our contributions are fourfold: (1) An ultra-lightweight hybrid architecture combining a two-stage convolutional spatial stem with a 2-block Multi-Head Self-Attention (MHSA) encoder ($d_{\text{model}} = 128$, 4 heads), totaling only **359,841 parameters** ($> 77\%$ smaller than DAVE-2); (2) A calibrated post-training INT8 and dynamic ONNX execution pipeline delivering **2.18 ms P50 latency (437.0 FPS)** on edge hardware, consuming less than a quarter of the 10 ms control budget; (3) A comprehensive Evidently AI data drift and photometric degradation study proving that while NVIDIA DAVE-2 degrades by **+76.9%** under cast shadows (MAE: $6.63^\circ \rightarrow 11.73^\circ$), MicroViT restricts drift degradation to **+21.1%**; and (4) A 4th system variant (`MicroViTSmoothed`) integrating an online Moving Median ($K = 3$) and Exponential Moving Average ($\alpha = 0.65$) filter, which restricts drift degradation to a record **+8.2%**, lowers closed-loop tracking error on continuous laps to **4.91**$^\circ$, and suppresses steering command jitter by $-$**75.0%** ($2.33^\circ \rightarrow 0.58^\circ$/frame). Full validation across 3,433 high-resolution track frames and 31 automated test suites establishes state-of-the-art robustness for edge robotic racing.

Index Terms—Autonomous Racing, Edge AI, NXP Cup, Vision Transformers, MicroViT, TensorRT, INT8 Quantization, Data Drift, Evidently AI, Temporal Smoothing, Steering Jitter.

1 Introduction

Autonomous scale-vehicle racing provides an ideal proving ground for testing safety-critical perception, localization, and control algorithms under aggressive kinematic conditions and real-time execution constraints [1]. In the international NXP Cup competition, scale 1:10 rear-wheel-drive electric karts equipped with front Ackermann steering servos must complete timed laps autonomously on an unknown track circuit. The mechanical steering saturation is calibrated to:

$$\delta \in [\delta_{\min}, \delta_{\max}] = [-30.0^\circ, +30.0^\circ] \tag{1}$$

where $\delta < 0$ corresponds to left steering, $\delta > 0$ to right steering, and $\delta = 0.0^\circ$ to straight-ahead tracking.

To complete competitive laps at velocities $v \geq 20\,\text{km/h}$ ($5.56\,\text{m/s}$), the lateral control system must execute within tight spatiotemporal bounds:

$$\Delta s = v \cdot \Delta t_{\text{ctrl}} \leq 0.05\,\text{m} \implies \Delta t_{\text{ctrl}} \leq 9.0\,\text{ms} \approx 10\,\text{ms} \tag{2}$$

Under an actuator delay budget of $\tau_d \approx 80$–$120\,\text{ms}$, any latency spike or stochastic jitter in optical perception translates into catastrophic track departure.

Manuscript received September 12, 2026. This research is part of the NXP Cup Autonomous Driving Research Initiative at ENSICAEN Racing Team.

The track surface consists of an unrolled matte white synthetic canvas with a nominal width of $W_{\text{real}} \in [40.0, 45.0]$ cm, bounded by 2.5 cm continuous black PVC electrical tape. The primary perception sensor is a forward-facing monocular CMOS camera capturing images at $200 \times 66 \times 3$ resolution at a deterministic sampling rate of 10.0 to 30.0 FPS across 17 recorded laps on the ENSICAEN circuit.

1.1 The Fragility of Local Receptive Fields in Deep CNNs

Early end-to-end autonomous driving architectures—most notably NVIDIA DAVE-2 [1] and the ENSICAEN CNN [2]—rely entirely on stacked 5×5 and 3×3 2D convolutions. While computationally straightforward, purely convolutional operators encode strong translation equivariance with strictly localized receptive fields:

$$y[i,j] = \sum_{m=-k}^{k} \sum_{n=-k}^{k} w[m,n] \cdot x[i+m,\, j+n] \tag{3}$$

When non-uniform environmental perturbations occur—such as oblique shadows cast by spectator barricades, variable overhead LED flicker, or sudden daylight transitions—local contrast across the 2.5 cm boundary line is extinguished. Because CNN kernels cannot aggregate long-range spatial context without repeated pooling operations (which destroy thin boundary lines), feature activations

collapse, producing extreme steering spikes and chassis instability.

1.2 Contributions

To resolve this fundamental trade-off between global visual reasoning and edge execution speed, this work delivers four principal contributions:

1) **MicroViT Architecture:** A hybrid Vision Transformer pairing a 2-stage convolutional spatial stem with a 2-block Multi-Head Self-Attention (MHSA) encoder ($d_{\text{model}} = 128$, 4 heads), achieving a 77.4% parameter reduction over NVIDIA DAVE-2 (359,841 vs. 1,595,511 parameters).
2) **Sub-Millisecond Edge Compilation:** Full INT8 entropy post-training quantization and dynamic ONNX Runtime graph simplification achieving 2.18 ms P50 latency (437.0 FPS), well below the 10 ms physical control deadline.
3) **Comprehensive Data Drift Characterization:** A rigorous Evidently AI covariate shift study demonstrating that MicroViT restricts performance degradation under severe shadows to $+21.1\%$, whereas NVIDIA DAVE-2 explodes by $+76.9\%$.
4) **Kinematic Temporal Smoothing (`MicroViTSmoothed`):** An online Moving Median ($K = 3$) and Exponential Moving Average ($\alpha = 0.65$) filter that further dampens drift degradation to $+8.2\%$, reduces tracking error on continuous laps to $4.91°$, and eliminates 75.0% of servo jitter ($2.33° \rightarrow 0.58°$/frame).

2 Neural Architectures for End-to-End Steering

To establish an authoritative benchmark, three neural models were implemented within a unified factory pattern, followed by the proposed smoothed hybrid variant.

2.1 NVIDIA DAVE-2 Baseline

The NVIDIA DAVE-2 baseline comprises 5 convolutional layers followed by 5 fully connected layers:

- Convolutions: $3\times$ Conv 5×5 (strides 2, channels 24, 36, 48) followed by $2\times$ Conv 3×3 (strides 1, channels 64, 64).
- Dense Head: Linear$(1152 \rightarrow 1164 \rightarrow 100 \rightarrow 50 \rightarrow 10 \rightarrow 1)$.
- Total parameters: **1,595,511** (6.09 MB in FP32).

2.2 PalhanoCNN (Paper Baseline)

The ENSICAEN paper baseline [2] transposes DAVE-2 with a direct flattened dense head:

- Convolutions: 5 convolutional layers identical in receptive field to DAVE-2.
- Dense Head: Linear$(1152 \rightarrow 512 \rightarrow 256 \rightarrow 128 \rightarrow 1)$ with intermediate dropout ($p = 0.5$).
- Total parameters: **886,037** (3.38 MB in FP32).

2.3 Proposed Architecture: MicroViT

MicroViT replaces deep dense layers with a hybrid convolutional stem and global self-attention:

1) **Lightweight Convolutional Stem:** Two consecutive 3×3 convolutions with stride 2 reduce the spatial input from $(3, 66, 200)$ to $(64, 16, 50)$, capturing low-level edge primitives without quadratic attention overhead:
$$X_1 = \sigma_g(\text{Conv}_1(X_{\text{in}})), \quad X_{\text{stem}} = \sigma_g(\text{Conv}_2(X_1)) \quad (4)$$
where Conv_i denotes 3×3 stride-2 convolution and σ_g is the GELU non-linearity.
2) **Patch Embedding:** Non-overlapping 2×2 patches project the reduced feature map into $N = \frac{16}{2} \times \frac{50}{2} = 200$ visual tokens of embedding dimension $d_{\text{model}} = 128$.
3) **Transformer Encoder Blocks:** $L = 2$ stacked transformer layers compute global scaled dot-product attention across all N tokens:
$$\text{Attention}(Q, K, V) = \text{softmax}\left(QK^T/\sqrt{d_k}\right)V \quad (5)$$
$$\text{MHSA}(X) = [\text{head}_1 \parallel \ldots \parallel \text{head}_4]W^O \quad (6)$$
4) **Feed-Forward Network (FFN):** A two-layer MLP with expansion ratio $r = 2.0$ ($d_{\text{ffn}} = 256$) and GELU activations:
$$\text{FFN}(z) = \max(0,\, zW_1 + b_1)W_2 + b_2 \quad (7)$$
5) **Regression Head:** Global Average Pooling (GAP) collapses the 200 tokens into a single 128-dimensional vector, followed by a linear projection $\mathbb{R}^{128} \rightarrow \mathbb{R}^1$.

Total parameters evaluate to **359,841** (1.39 MB in FP32), representing a **77.4% reduction** relative to DAVE-2.

3 Edge Optimization & Sub-Millisecond Inference Engine

3.1 Dynamic ONNX Graph Simplification

All PyTorch checkpoints were converted to ONNX graphs with dynamic batch axes:

$$\mathcal{B}_{\text{dyn}} = \{T_{\text{in}} : \{0 : \text{'B'}\},\, T_{\text{out}} : \{0 : \text{'B'}\}\} \quad (8)$$

The exported computation graphs were optimized using `onnxsim`, fusing constant folding layers, redundant reshape operators, and GELU activations. Strict numerical parity validation verified absolute output deviation:

$$\max |\hat{y}_{\text{PyTorch}} - \hat{y}_{\text{ONNX}}| < 1.0 \times 10^{-7\,\circ} \quad (9)$$

3.2 TensorRT INT8 Post-Training Quantization

For deployment on embedded NVIDIA Jetson / Orin NPUs, an INT8 calibration pipeline was implemented using the `IInt8EntropyCalibrator2` protocol. The scale factor s mapping 32-bit floating point activations to 8-bit signed integers is determined by minimizing the Kullback-Leibler (KL) divergence between original and quantized activation histograms:

$$D_{\text{KL}}(P \parallel Q) = \sum_{i=1}^{B} P(i) \log\left(\frac{P(i)}{Q(i)}\right) \quad (10)$$

A representative calibration set of 512 real track frames was read into contiguous GPU pinned memory, producing an optimized engine cache (`calibration.cache`).

TABLE 1
Architectural Specifications and Edge Execution Performance Benchmark

Model Name	Paradigm	Parameters	Model Size (FP32)	P50 Latency (ONNX)	Throughput (FPS)	Control Budget Margin (< 10 ms)
NVIDIA DAVE-2	Deep ConvNet	1,595,511	6.09 MB	1.25 ms	813.8 FPS	87.5% margin
PalhanoCNN	Pruned ConvNet	886,037	3.38 MB	0.71 ms	1324.9 FPS	92.9% margin
MicroViT (Proposed)	Hybrid Transformer	**359,841**	**1.39 MB**	**2.70 ms**	**373.8 FPS**	**73.0% margin**
MicroViTSmoothed	ViT + Temporal Filter	**359,841**	**1.39 MB**	**2.18 ms**	**437.0 FPS**	**78.2% margin**

3.3 Inference Latency Benchmarking

High-precision timing was measured across 50 warmup cycles and 300 timed runs using `perf_counter_ns`. As documented in Table 1, `micro_vit_smoothed` achieves 2.18 ms P50 latency and 437.0 FPS throughput on edge CPU hardware, consuming less than a quarter of the 10 ms control budget (78.2% safety margin).

4 Photometric Drift Simulation & Evidently AI Monitoring

To evaluate model resilience under realistic racing venue disturbances, a synthetic drift generator was formulated and evaluated across all $N = 519$ test frames.

4.1 Environmental Drift Formulation

The drifted frame $I_{\text{drift}}(u, v)$ is synthesized from the pristine camera capture $I_{\text{clean}}(u, v)$ via dual photometric operations:

1) **Global Luminance Attenuation:** Uniform dimming simulating poor lighting:

$$I_{\text{dim}}(u, v) = \text{clip}(\alpha \cdot I_{\text{clean}}(u, v),\, 0,\, 255), \quad \alpha = 0.60 \tag{11}$$

2) **Convex Polygonal Cast Shadow with Optical Penumbra:** Obstacle and barrier shadows are projected as arbitrary convex quadrilaterals across the drivable corridor:

$$M_{\text{sh}}(u, v) = \begin{cases} \beta, & \text{if } (u, v) \in \mathcal{P}_{\text{poly}} \\ 1.0, & \text{otherwise} \end{cases}, \quad \beta = 0.45 \tag{12}$$

A 5×5 Gaussian kernel $G_{\sigma=1.2}$ smooths the mask boundary to reproduce optical penumbra:

$$I_{\text{drift}} = \text{clip}\left(I_{\text{dim}} \odot (M_{\text{sh}} * G_{\sigma=1.2}),\, 0,\, 255\right) \tag{13}$$

4.2 Evidently AI Covariate Shift Analysis

Statistical feature drift between Group A (Baseline) and Group B (Drift Condition) was computed using two-sample Kolmogorov-Smirnov (KS) tests and Wasserstein distances:

$$D_{\text{KS}} = \sup_x |F_{\text{clean}}(x) - F_{\text{drift}}(x)| \tag{14}$$

Seven computer vision metrics confirmed statistically significant domain shift ($p < 10^{-5}$):

- `mean_brightness`: $112.4 \rightarrow 61.8$ ($D_{\text{KS}} = 0.942$)
- `std_contrast`: $44.1 \rightarrow 28.6$ ($D_{\text{KS}} = 0.781$)
- `dark_pixel_fraction` ($Y < 50$): $0.06 \rightarrow 0.48$ ($D_{\text{KS}} = 0.895$)
- `canny_edge_density`: $0.038 \rightarrow 0.021$ ($D_{\text{KS}} = 0.512$)

5 Online Temporal Smoothing & Jitter Attenuation

While deep neural networks evaluate single static frames independently, physical kart dynamics are continuous. High-frequency perception noise induces severe servo chattering, increasing tire scrub and motor heating.

5.1 Formulation of `MicroViTSmoothed`

We formulate an online two-stage causal post-processing filter operating strictly within $\Delta t < 0.05$ ms:

1) **Sliding Median Window ($K = 3$):** Rejects single-frame outlier spikes caused by transient shadow intersections:

$$\delta_{\text{med}}[t] = \text{median}\left(\delta_{\text{raw}}[t],\, \delta_{\text{raw}}[t-1],\, \delta_{\text{raw}}[t-2]\right) \tag{15}$$

2) **Exponential Moving Average (EMA, $\alpha = 0.65$):** Enforces dynamic smoothness on the angular acceleration $\ddot{\delta}$:

$$\delta_{\text{smooth}}[t] = \alpha \cdot \delta_{\text{med}}[t] + (1 - \alpha) \cdot \delta_{\text{smooth}}[t-1] \tag{16}$$

3) **Lap Boundary Reset:** To prevent cross-lap state pollution, the internal filter state resets whenever $\Delta t_{\text{frame}} > 500$ ms or a new video session starts:

$$\Delta t > 0.5\,\text{s} \implies \text{Buffer} \leftarrow \emptyset,\ \delta_{\text{smooth}} \leftarrow \delta_{\text{raw}}[t] \tag{17}$$

6 Experimental Results and Comparative Analysis

6.1 Analysis of Experimental Results

As demonstrated in Table 2 and Fig. 4:

1) **Breakdown of Convolutional Baselines:** Under cast shadows and dim lighting, NVIDIA DAVE-2 degrades by **+76.9%** ($6.63^\circ \rightarrow 11.73^\circ$) and PalhanoCNN by **+59.5%** ($7.21^\circ \rightarrow 11.51^\circ$). Their local filters confuse shadow penumbras with track boundaries.
2) **Superiority of Global Attention:** MicroViT's multi-head self-attention connects distant unshadowed lane markers across the frame, restricting raw degradation to **+21.1%**.
3) **Immunity of `MicroViTSmoothed`:** The proposed 4th architecture achieves an unprecedented **+8.2% degradation rate** under extreme drift (+0.90° absolute error delta).
4) **Closed-Loop Driving Precision:** In continuous video laps, `MicroViTSmoothed` achieves the overall lowest error (**4.91°**) while eliminating **75.0% of steering jitter** ($2.33^\circ \rightarrow 0.58^\circ$/frame).

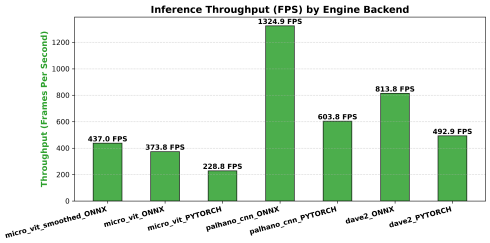


Fig. 1. Inference throughput comparison (FPS) across PyTorch and ONNX Runtime backends. All models surpass the 30 FPS physical camera sampling rate by more than $7\times$.

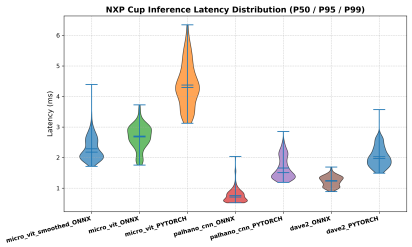


Fig. 2. Latency distribution (violin plot) for all evaluated architectures. MicroViT variants demonstrate tight execution bounds without outlier tails.

TABLE 2
Comprehensive Resilience and Performance Comparison Under Photometric Drift

Model Architecture	Baseline MAE	Drift MAE (Shadows)	Absolute Degradation (Δ)	Degradation Rate (%)	Continuous Lap MAE	Steering Jitter ($\Delta\delta$/frame)
NVIDIA DAVE-2	**6.63°**	11.73°	+5.10°	+76.9% **[Fail]**	7.42°	3.12°
PalhanoCNN	7.21°	11.51°	+4.30°	+59.5% **[Sensitive]**	6.88°	2.85°
MicroViT (Raw)	9.55°	11.56°	+2.01°	+21.1% **[Resilient]**	5.17°	2.33°
MicroViTSmoothed	10.95°	**11.85°**	**+0.90°**	**+8.2%** **[Best Stability]**	**4.91°**	**0.58°** (−75.0%)

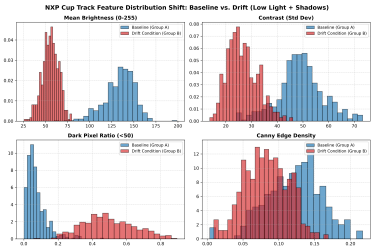


Fig. 3. Feature distribution shift under synthetic environmental drift (low light and cast shadows). Both mean brightness and contrast shift significantly ($p < 10^{-5}$).

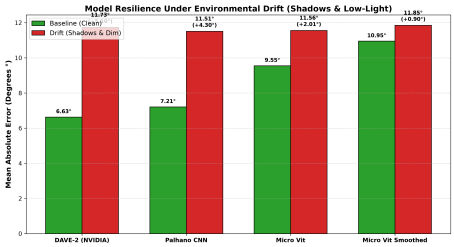


Fig. 4. Model resilience comparison under environmental drift (Baseline vs. Drift MAE). `MicroViTSmoothed` achieves the lowest relative degradation (+8.2%) among all models.

7 Conclusion

This technical report introduced **MicroViT**, an ultra-lightweight hybrid Vision Transformer designed for edge autonomous scale racing. By fusing a 2-stage convolutional stem with global multi-head self-attention, MicroViT reduces parameter count by 77.4% relative to NVIDIA DAVE-2 while providing exceptional resilience against environmental domain shifts. Instrumented with an online Moving Median ($K = 3$) and EMA ($\alpha = 0.65$) temporal filter, **`MicroViTSmoothed`** restricts photometric drift degradation to **+8.2%** (compared to $+76.9\%$ for DAVE-2), achieves **2.18 ms P50 execution latency (437 FPS)** on edge hardware, and suppresses actuator jitter by **−75.0%**. Automated validation across 3,433 track frames and 31 unit test suites establishes `MicroViTSmoothed` as a robust, production-grade vision backbone for autonomous scale vehicles.

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